

# Khush Shah

Carnegie Mellon University | B.S. in Computer Science & Mathematics | Expected May 2029  
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Fully work authorized | U.S. Citizen

## EDUCATION

### Carnegie Mellon University

*B.S. in Computer Science & Mathematics*

Pittsburgh, PA

*Expected May 2029*

- **Relevant Classes:** Deep Learning (Graduate Level), Machine Learning, Data Structures & Algorithms, Competition Programming, Computer Systems, Linear Algebra, Multivariable Calculus, Differential Equations

## EXPERIENCE

### Maximor

*Software Engineer Intern*

June 2026 – Present

*June 2026 – August 2026*

- Designed and deployed the company's first multi-agent system on Claude Managed Agents: a coordinator agent delegating to subagents via cross-thread tool-call routing, replacing separate Temporal workflows
- Built the human-in-the-loop clarification system for agents in production: answer/skip/rewind lifecycle, reviewer UI, and a managed-agent runtime that pauses live sessions, persists them durably, and resumes via Temporal
- Served as a forward-deployed engineer on a ~8-person team for a \$1M client, shipping requested features and fixes

*Software Engineer (Part-Time)*

*August 2026 – Present*

- Revising and extending the production fleet of Claude Managed Agents: rewriting system prompts, tightening tool schemas, reworking input framing, and debugging live agent runs to improve reliability and output quality
- Designing the company's agent evaluation harness: regression evals gating the migration of live customer workloads to newer models, plus orchestration tooling to run agents reproducibly

### Carnegie AI Safety Initiative (CASI)

September 2025 – Present

- Studied transformers in depth through a technical AI safety reading group, covering mechanistic interpretability, RLHF, alignment techniques, and the fundamentals of modern ML research on LLM internals
- Co-leading a 3-person research project on AI safety alignment robustness, producing empirical research

### ScottyLabs Tech Team | CMU Courses Backend

September 2025 – June 2026

- Built automated Rust system to query CMU APIs and recursively discover related course resources
- Wrote Python ETL to normalize course data into Postgres and expose it via JSON and CSV endpoints
- Rebuilt the web app in TypeScript + React using TanStack Router; platform used by 1000s of CMU students

## RESEARCH

### Geometric Framework for Predicting Fragility of LLM Safety Training Methods

April 2026 – Present

- Using RepE-extracted concept vectors and Hessian eigenvector analysis to show safety concepts occupy lower-rank, higher-curvature subspaces than capability directions
- Showing that benign fine-tuning updates preferentially project onto fragile safety directions, with representational rank and layer depth as independent fragility predictors
- Building a framework that predicts DPO- vs PPO-trained models' alignment robustness from geometric properties, validating on HarmBench across Llama, Mistral, and Qwen; targeting ICLR 2027 workshops

## LEADERSHIP EXPERIENCE

### Los Altos Hacks | Organizer & Sponsorship Lead

2021 – 2025

- Organized the world's largest high-school hackathon; personally raised \$20,000+ in sponsorships; gave speeches to 1000+ students and mentored many new members

### Computer Engineers of the Next Generation (CENG) | Senior VP

2022 – 2025

- Managed 100+ volunteers teaching 24+ coding classes (Python, Java, JavaScript) to underserved schools; created over \$1M of impact; earned the *Presidential Volunteer Service Award* for 200+ service hours in a single year

## TECHNICAL SKILLS

**ML:** Claude Managed Agents, LLM APIs, MCP; implemented transformers from scratch; completed ARENA AI safety curriculum; working knowledge of LLM internals, attention, RLHF, and alignment techniques

**Languages:** Python, Java, C++, C, C#, JavaScript, TypeScript, Rust, SQL, CSS, Standard ML,  $\text{\LaTeX}$

**Tools & Platforms:** Git, Linux, Docker, PostgreSQL, Snowflake, Temporal, Firebase, MongoDB, GCP, AWS

**Libraries & Frameworks:** React, Node.js, FastAPI, PyTorch, TensorFlow, NumPy, pandas, scikit-learn